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RESEARCH ARTICLE

Rational Design of Robust and Efficient Natural Leather-Based Ionic Thermoelectric Detectors for Energy-Autonomous and Anti-scalding

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Abstract

Compared with those traditional initiating devices of anti-scalding systems, ionic thermoelectric sensors with energy-autonomous performance show higher reliability. However, the current ionic thermoelectric materials (i-TEs) suffer from complex nano-/micro-channel design, high production costs, environmentally unfriendly, weak mechanical properties, as well as the low moving speed of ions. Herein, the functional leather collagen fibers-bearing natural channels are employed as the polymer matrixes, while the trisodium citrate (SC) organic acid salt exhibits the function of cationic moving self-enhancement as the primary mobile ions for signaling. Including numerous and suitable nano-/micro-channels together with fast-moving cations, the leather-based i-TEs (LITE), LITE-$SC_{0.75\ M}$, possess excellent thermoelectric properties, achieving a Seebeck coefficient of 6.23 mV/K, a figure of merit of 0.084, and an energy conversion efficiency of 2.12%. Combined with its excellent thermal stability, mechanical performance, flexibility, durability, low cost, and outstanding capabilities for low-grade heat harvesting and thermal sensing, the LITE-$SC_{0.75\ M}$ detector bearing long service life would show great promise in automatic anti-scalding alarm suitable for multiple scenarios and extreme environments. Therefore, the present work aims to design an efficient, robust, and energy-autonomous leather collagen fibers-based thermoelectric detector to address the limitation of current anti-scalding alarm technology as well as drive advancements in the nano-energy and its effective conversion field.

Keywords Energy conversion · Ionic thermoelectricity · Functional leather collagen fibers · Flexible sensor · Anti-scalding alarm

1 Introduction

In our fast-moving modern world, safety remains a top priority for humans. Among the essential safety measures is installing automatic anti-scalding systems. These systems play a vital role in warning occupants of potential dangers, and even potentially saving lives. The initiating devices commonly refer to heat/temperature detectors, traditionally including thermocouples, thermistors, and infrared sensors [1–3]. Although effective and widespread use, they all need an external power supply, which is undoubtedly short of reliability considering the ease of failure of the power supply in some dangerous situations. Therefore, it is particularly important to explore new and power-autonomous anti-scalding initiating technology to safeguard our lives and property.

Ionic thermoelectric nanotechnology, the direct conversion of heat into electricity based on the Seebeck effect, has garnered significant attention in the realm of sustainable

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energy development [4–6]. It is worth noting that this technology shows great promise for anti-scalding applications due to the temperature-sensitive nature of its material. More importantly, the ionic thermoelectric sensors are energy-autonomous. When the sensor comes into contact with a high-temperature object, it becomes self-powered and activates a danger warning. This enhances its reliability compared to traditional anti-scalding system initiators. To achieve this goal, the design and fabrication of the corresponding ionic thermoelectric materials (i-TEs) are crucial.

As for i-TEs, especially the quasi-solid state i-TEs, they mainly contain two components, polymer matrixes or substrates providing nano-/micro-channel and mobile ions for signaling. So far, multitudes of polymer matrixes and mobile ions have been tried to assemble and fabricate functional i-TEs. For example, the ionic liquids (EMIM:BF_4, BMIM:Cl, EMIM:DCA) [7–9], strong bases (NaOH, KOH) [10–12], strong acidic and bases salts (NaCl, KCl, NaTFSI) [13–15], or redox ion pairs (Fe^{3+}/Fe^{2+}, $[Fe(CN)_6]^{3-}/[Fe(CN)_6]^{4-}$, I^-/I_3^-) [16–18] were commonly used to provide mobile ions. However, some challenges still exist to impede their widespread applications. (i) The vast majority of ionic liquids are expensive and potentially hazardous to the ecosystem and human health. (ii) The strong bases/acids or their salts commonly demonstrate low moving speeds in the nano-/micro-channel thus weak thermoelectric performance. (iii) Only the redox ion pairs have difficulty in increasing Seebeck coefficients due to entropy difference limitations. On the other hand, the polymer matrix includes synthetic or natural polymers. The synthetic polymer matrixes often needed material modification [19–21], directional freezing [22–24], and stretch-induced crystallization [25] strategies to endow them with suitable channel architectures for ion mobility. However, such complex design and synthesis showed tedious work-up procedures, time-consuming, high production costs, and even environmentally unfriendly, thereby limiting their potential applicability. Moreover, natural polymers with inherent nano-/micro-channel structures often suffer from weak mechanical properties and flexibility, restricting their service life and usage scenarios. Therefore, to realize the ionic thermoelectric technology in initiating the anti-scalding, rational design of robust, efficient, green, and low-cost quasi-solid state i-TEs is significantly important.

Leathermaking is an industry that transforms the corruptible into the mysterious. Based on its distinct structure and performance, the development of leather collagen fibers as the polymer matrixes of quasi-solid state i-TEs is promising. First of all, leather collagen fibers demonstrate multilayered structures, which endow abundant and inherent flexible nano-/micro-channels available for ions flowing, the key requirement for thermal-induced electricity material. Second, leather collagen fibers contain abundant functional groups, the interface structure attributes (electro-negativity or positivity) of which could be adjusted optionally in compliance with the mobile ions, thus extending their serviceable range. Third, the leather collagen fibers were interwoven tightly into a well-organized three-dimensional network, providing exceptional mechanical properties, hydrothermal stability, flexibility, durability, and corrosion resistance. Furthermore, leather matrixes show abundant natural sources, biocompatibility, and degradability, resulting in environmental friendliness, low cost, and good applicability.

On the other hand, organic acid salts, generated by the neutralization of organic acids and alkaloids that are widely distributed in plants, agricultural by-products, or animal tissues, have the advantages of biocompatibility, environmental friendliness, as well as inexpensiveness [26–28]. In a polymer matrix environment, certain organic acid salts can form hydrogen bonds between their acidic anions (*e.g.*, $-COO^-$ and $-O^-$) and the functional groups of the polymer matrix, such as -COOH, $-NH_2$, and -OH found in leather collagen. This interaction theoretically facilitates the movement of the corresponding salt cations under thermal stimulation.

Based on these, herein, natural leather collagen fibers were employed to act as the polymer matrixes and trisodium citrate organic acid salt as the primary mobile ions, the product of which demonstrated robust structure and efficiently energy-autonomous, promising in automatic anti-scalding. To achieve this, pickled sheep skin was employed as raw material, which underwent an *Emblic* and glutaraldehyde (GTA) tanning process (labeled as $\text{Leather}_{Emblic/GTA}$). Subsequently, the $\text{Leather}_{Emblic/GTA}$ was soaked in an aqueous solution containing not only trisodium citrate (SC) but also $K_3[Fe(CN)_6]/K_4[Fe(CN)_6]$ to exert thermogalvanic and thermodiffusive synergistic effects. Further optimizing the mobile ions of SC and nano-/micro-channels of leather matrixes, the final product [leather-based i-TEs (LITE)], $\text{LITE-SC}_{0.75\ M}$, achieved satisfactory thermoelectric properties. The mechanism of efficient thermo-induced electricity of $\text{LITE-SC}_{0.75\ M}$ was attempted to be deciphered by technical characterization. After evaluation of thermal stability, mechanical properties, durability, low-grade heat harvesting, and heat-sensing capability, the potential application of $\text{LITE-SC}_{0.75\ M}$ in automatic anti-scalding was assessed. Hence, this research intends to offer a robust and efficient natural leather-based ionic thermoelectric detector with energy-autonomous performance to break through the dilemma of current anti-scalding alarm technology.

2 Experimental Section

2.1 Materials

The pickled sheep skin was provided by Sichuan ShiJi Leather Co., Ltd. (China). Vegetable tannin *Emblic* was obtained from Guangxi Fusui Shengli Glue Co., Ltd. (China). Sodium chloride (NaCl, AR, ≥ 99.5%), sodium bicarbonate ($NaHCO_3$, AR, ≥ 99.5%), potassium ferrocyanide ($K_4[Fe(CN)_6 \cdot 3H_2O]$, AR, ≥ 99.5%), trisodium citrate dihydrate ($Na_3C_6H_5O_7 \cdot 2H_2O$, AR, ≥ 99.0%), sodium acetate anhydrous (CH_3COONa, AR, ≥ 99.0%), sodium formate dihydrate ($HCOONa \cdot 2H_2O$, AR, ≥ 99.5%), potassium chloride (KCl, AR, ≥ 99.5%), sodium sulfate anhydrous (Na_2SO_4, AR, ≥ 99.0%), and sodium oxalate ($Na_2C_2O_4$, AR, ≥ 99.8%) were obtained from Tianjin Damao Chemical Reagent Co., Ltd. (China). Formic acid (HCOOH, AR, ≥ 99.5%), glutaraldehyde solution ($C_5H_8O_2$, AR, ≥ 25.0%), and potassium sodium tartrate tetrahydrate ($NaKC_4H_4O_6 \cdot 4H_2O$, AR, ≥ 99.0%) were provided by Tianjin Kemiou Chemical Reagent Co., Ltd. (China). Potassium hexacyanoferrate ($K_3[Fe(CN)_6 \cdot 3H_2O]$, AR, ≥ 99.5%) was purchased from Tianjin Opsheng Chemical Co., Ltd. (China). Lithium chloride anhydrous (LiCl, AR, ≥ 99.0%) was purchased from Shanghai Macklin Biochemical Technology Co., Ltd. (China). Sodium nitrate ($NaNO_3$, AR, ≥ 99.5%) was obtained from Chengdu Kelong Chemical Co., Ltd. (China). The above reagents were employed without additional purification treatment.

2.2 Fabrication of the Leather-Based Ionic Thermoelectricity (LITE-SC_x)

The leather-based ionic thermoelectricity (LITE-SC_x) was fabricated in two steps: first preparing the $Leather_{Emblic/GTA}$ by tanning the pickled skin and then introducing ions to prepare LITE-SC_x, where x denoted the concentration of the $Na_3C_6H_5O_7$ (SC) solution, including 0, 0.25, 0.5, 0.75, 1.0, and 1.5 mol/L. The detailed fabrication process and parameters could be referred to the Supplementary Information (Procedure S1, Fig. S1, and Table S1).

2.3 Thermoelectric performance test

To test the ionic thermoelectric property of LITE-SC_x, Ag electrodes were used to completely cover the upper and lower surfaces of the LITE-SC_x. A SourceMeter (Keithley2450, US) was connected to detect the potential difference. Two semiconductor Heating/Cooling Chips (TEC1-12,706, China) were connected to a Programmable Linear DC Power Supply (RIGOL DP832, China) to assemble a test device that can maintain a stable temperature difference (Fig. S2). A Multi-Channel Temperature/Data Recorder (Tonghui TH2552, China) was used to record real-time temperatures of the two ends of the LITE-SC_x. The open-circuit voltage of the LITE-SC_x was tested at temperature differences of 5, 10, 15, and 20 K with turn-on/off cycling at 150 s intervals, respectively. The ionic conductivity of the LITE-$SC_{0.75\ M}$ with different water contents (W = 60%, 50%, 40%, 30%) was analyzed by Electrochemical Workstation (Chenhua CHI600E, China). The thermal conductivity of the LITE-$SC_{0.75\ M}$ with different water contents was analyzed by a Thermal Constant analyser (Hot Disk TPS2200, Sweden).

2.4 Structural Characterization

The surface morphology, pore structure, elemental distribution, and chemical bonding of the LITE-$SC_{0.75\ M}$ were characterized. The cross-section of the platinum-sprayed lyophilized sample was analyzed by Field-Emission Scanning Electron Microscope (FE-SEM, ZEISS Gemini 360, Germany). The surface morphology was observed by an Ultra-Depth Three-Dimensional Microscope (UD-3DM, HIROX KH-8700, Japan). The pore size distribution was obtained by Mercury Piezometer (MP, AutoPore IV 9500, Micromeritics, US). The elemental distribution was analyzed by Energy-Dispersive Spectroscopy (EDS, Oxford Instruments, UK). The chemical bonding was disclosed by X-ray Photoelectron Spectroscopy (XPS, AXIS SUPRA, UK) and Fourier Transform Infrared Spectroscopy (FTIR, Nicolet 5700, US).

2.5 Functional Application of LITE-$SC_{0.75\ M}$ as an Automatic Anti-scalding Alarm Detector

Before functional application, the thermal stability and mechanical properties of LITE-$SC_{0.75\ M}$ were first evaluated (Procedure S2, Supplementary Information). Using Cu electrodes and in series with ten LITE-$SC_{0.75\ M}$, the heat harvesting device was constructed to harvest thermal energy from laptops, smartphones, forearms, and water, respectively. A simple heat-sensing glove was constructed by fixing five LITE-$SC_{0.75\ M}$ to each finger. Combining the low-grade heat harvesting and heat-sensing properties of LITE-$SC_{0.75\ M}$, the automatic anti-scalding alarm device was built. The device featured a sandwich structure, with LITE-$SC_{0.75\ M}$ placed in the center, an outer layer made of Cu as the thermal detection electrode, and a top layer made of biaxially oriented polypropylene film (BOPP) for flexible encapsulation. When the device was contacted with thermal sources, it could utilize the temperature difference between the touched object and thermal sources to perform thermoelectric conversion, thereby driving a signal transmitter.

3 Results and Discussion

3.1 Design and Fabrication of LITE-SC_x

Tanning that transfers the raw animal skin into mature leather is a vital and dominant process in leather making. The chemicals used in the tanning process mainly include vegetable tannins, metallic minerals, and syntans. In this work, the typical vegetable tannin, *Emblic*, was selected elaborately considering the following advantages: (i) vegetable tannins that widely exist in the leaves, barks, and fruits of plants demonstrate good biodegradability, environmental friendliness, and low cost, the reason of which the vegetable-tanned leather has been used for more than 4000 years until now [29–31]; (ii) the vegetable-tanned leather maximally remains its breathability, manifesting as that abundant and natural nano-/micro-channels conducive to the movement of ions are largely preserved, the most important performance for i-TEs; (iii) the vegetable tannins generally appear light brown, thus endowed the tanned leather with brunet in color (Fig. S3), which is beneficial to enhance its thermoelectric property because of brunet color improving the photothermal absorption capacity; (iv) the vegetable tannins are plant polyphenols with plentiful oxygen-containing groups, beneficial for constructing a negative environment for cation (exemplified by Na^+ and K^+) motion. As demonstrated in Fig. 1a, the vegetable tannin, *Emblic*, permanently changed the structure and properties of animal skin to obtain useful leather by multiple hydrogen bonds bonding with collagen fibers. Moreover, the common supplementary tanning agent, glutaraldehyde (GTA), was added based on the following benefits: (i) on account of covalent bonds generating between GTA and collagen fibers (such as -NH_2 of Histidine) via Schiff-bases reaction (Fig. 1a), it can intensify the leather such as improving its mechanical property and thermal stability, guaranteeing its service life and usage scenarios as a detector for automatic anti-scalding alarm; (ii) the GTA tanning process would consume more cationic amino of collagen fibers, thus further imparting a negative environment for cation motion to improve its thermoelectric property. After tanning, the Leather$_{Emblic/GTA}$ was subject to simple soaking treatment by an aqueous solution including $K_3[Fe(CN)_6]/K_4[Fe(CN)_6]$ and SC, where the -OH and -COO^- groups of citrate were probably hydrogen-bonded to *Emblic* (Fig. 1a). By varying the concentration of the used SC, the final product was obtained as the leather-based ionic thermoelectric material (LITE-SC$_x$).

As mentioned above, this work intends to develop natural leather as the polymer matrixes of quasi-solid state i-TEs based on the fact that leather collagen fibers demonstrate multilayered structures, including collagen fibers, fibrils, and triple helixes, which endow abundant and inherent flexible channels available for ions flowing (Fig. 1b). Herein, due to the limitation of anion (citrate) movement, those cations including Na^+ and K^+ migrated along the triple-stranded helix from warmer to colder region to generate a concentration difference, thus potential difference, the mechanism of electricity induced by temperature difference (thermal diffusion). Meanwhile, as a common strategy [32–34], the reduction and oxidation reactions occurred at the interface of the warmer and colder electrodes by introducing $[Fe(CN)_6]^{4-}$ and $[Fe(CN)_6]^{3-}$ redox couple to synergize thermogalvanic with thermal diffusion effects. By rationally designing polymer matrixes and optimizing mobile ions, the product of robust and efficient natural leather-based ionic thermoelectric material was intended to assemble a thermal sensor, which could harvest low-grade heat as the stimulus, demonstrating a sensitive heating sensing ability (Fig. 1c). Moreover, by assembling the thermal sensor with Cu electrodes and flexible encapsulation using biaxially oriented polypropylene film (BOPP), the prepared ionic thermoelectric detector featuring as sandwich structure had the potential to realize energy-autonomous, providing a new perspective on breaking through the dilemma of current anti-scalding alarm technology.

To obtain high thermoelectric properties, first of all, the concentration of the used SC that affected the cation motion was optimized. Herein, the concentrations of SC including 0, 0.25, 0.5, 0.75, 1.0, and 1.5 mol/L were introduced. Figures 2a and Fig. S4a illustrate the open-circuit voltage (V_{oc}) curves of LITE-SC$_x$ under temperature difference (ΔT) of 5, 10, 15, and 20 K. Regardless of the effect of SC concentration, higher ΔT achieved higher V_{oc}, a normal phenomenon based on the law of conservation of energy (the more heat inputting, the more electrical outputting). With increasing the added SC, the change of V_{oc} for all ΔT went through two stages, first increase of V_{oc} and then decrease. Such variation trend could be further quantitative characterization by the Seebeck coefficient (*S*) (Fig. 2b and Fig. S4b), the major parameter measuring thermoelectric properties (the higher the coefficient, the better the thermoelectric performance). As the concentration of SC was 0, the Seebeck coefficient of LITE-SC$_{0\ M}$ (2.24 mV/K) was attributed to the thermogalvanic effect of $[Fe(CN)_6]^{4-}$ and $[Fe(CN)_6]^{3-}$ redox couple. By increasing the SC from 0 to 0.75 mol/L, the Seebeck coefficient values showed sustainable growth, while reaching a maximum of 6.23 mV/K of the LITE-SC$_{0.75\ M}$ sample. Such a phenomenon was tried to disclose as illustrated in Fig. 2c. The addition of more SC would introduce more cations Na^+ and anions citrate. The organic acidic anions citrate could form hydrogen bonding with functional groups of the polymer matrix (*Emblic* and leather collagen), thus facilitating the movement of the Na^+ under thermal stimulation, manifesting as the promoted Seebeck coefficient when increased the SC. However, when the concentration of SC further increased, exemplified by 1.0 and even 1.5 mol/L, the Seebeck coefficients decreased, 3.95 mV/K of LITE-SC$_{1\ M}$ and 3.74 mV/K of LITE-SC$_{1.5\ M}$. This phenomenon could be ascribed to the following reasons: (i) the high concentration of SC enhancing the interaction between ions, thus affecting the ionization balance and reducing the mobility of ions; (ii) the high concentration of citrate causing the accumulation of *Emblic*, thus blocking the channel, and finally restricting the

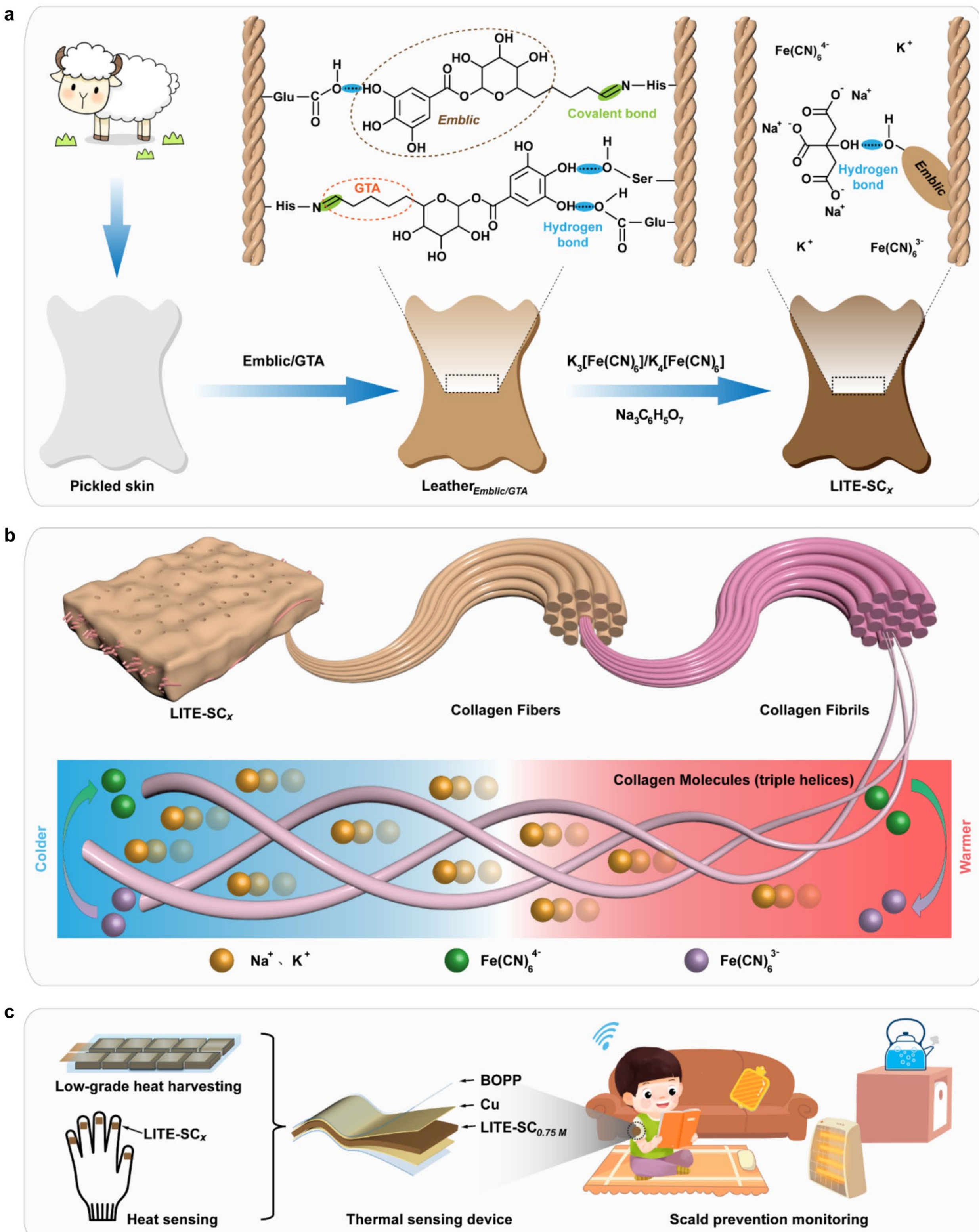

Fig. 1 **a** Schematic of the preparation and chemical bonding structure of leather-based ionic thermoelectricity (LITE-SC$_x$). **b** Schematic of the ionic thermoelectric principle based on thermodiffusion effect and thermogalvanic effect synergy. **c** Schematic illustration of the assembled LITE-SC$_x$ as the automatic anti-scalding alarm device

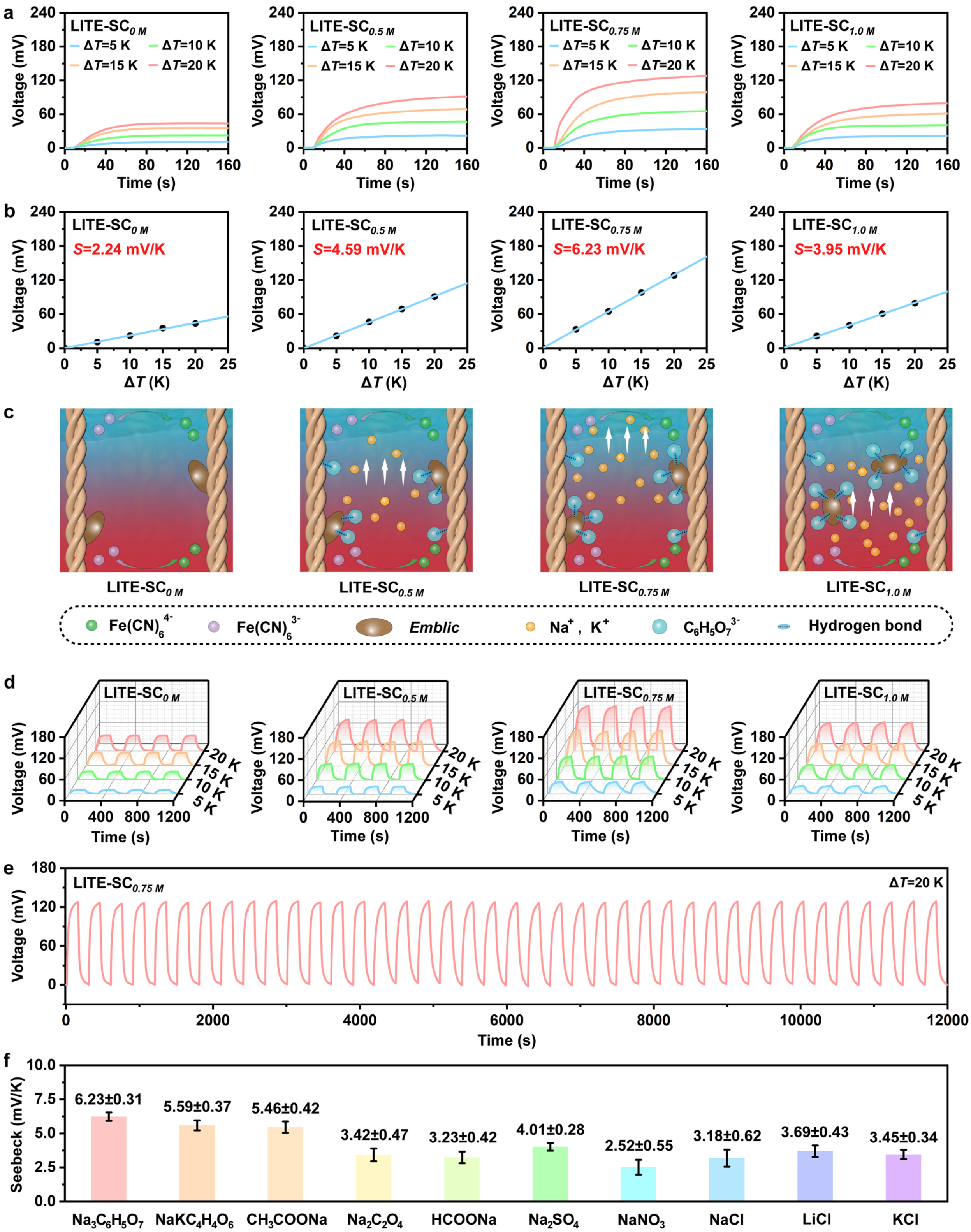

Fig. 2 **a** Open-circuit voltage (V_{oc}) curves of LITE-SC$_x$. **b** Seebeck coefficient of LITE-SC$_x$ obtained from linear fitting of voltage and ΔT. **c** Schematic of the thermoelectric property of LITE-SC$_x$ affected by SC with different concentrations. **d** V_{oc} curves of LITE-SC$_x$ at turn-on/off cycles. **e** V_{oc} curve of LITE-SC$_{0.75\ M}$ cycled continuously for 12,000 s under $\Delta T = 20$ K. **f** The comparison of Seebeck coefficients when doped different salts into leather-based ionic thermoelectric materials

free movement of cations (Fig. 2c); (iii) the high concentrations of citrate-*Emblic* migrating together and accumulating at the electrolyte/electrode interface, thus reducing the interfacial activity. Therefore, it could be concluded that the SC concentration of 0.75 mol/L was the best, the product of which (LITE-$SC_{0.75\ M}$) achieved the highest Seebeck coefficient. Moreover, the samples exhibited good stability of thermoelectric performance, as illustrated in Fig. 2d that repeatedly turned on/off ΔT. The time of turning on ΔT was set as 150 s as the voltage rise concentrated in the first 40 s and very slow growth at 150 s (Fig. 2a). The time of turning off ΔT was 150 s considering the V_{oc} of LITE-$SC_{0.75\ M}$ decay to about 0 mV within 150 s. The excellent stability and durability of the thermoelectric performance of LITE-$SC_{0.75\ M}$ could be proved by continually cyclic turning on/off ΔT. As could be seen in Fig. 2e, even after 12,000 s circulation, the maximum V_{oc} did not show any significant degradation, showing its application feasibility. When continuously turned on ΔT of 20 K for 4000 s, the final V_{oc} was stabilized at around 210 mV, indicating the potential of LITE-$SC_{0.75\ M}$ to generate high voltage (Fig. S5).

In comparison, other salts including the common organic acid salts ($NaKC_4H_4O_6$, CH_3COONa, $Na_2C_2O_4$, and HCOONa) and inorganic salts (Na_2SO_4, $NaNO_3$, NaCl, LiCl, and KCl) were introduced identically, respectively. To make the comparison feasible, the cation content of different salts remained consistent (*i.e.*, the cationic ratio of 1). For example, if the concentration of $Na_3C_6H_5O_7$ (SC) was 0.75 mol/L, the concentration of $Na_2C_2O_4$ was 1.125 mol/L, while the concentration of NaCl was 2.25 mol/L. Based on their values of the Seebeck coefficient (Fig. 2f), it could be determined that, on the one hand, the leather-based polymer matrix doped with organic acid salts, especially $NaKC_4H_4O_6$ and CH_3COONa achieved a superior Seebeck coefficient than those of inorganic salts, which proved the hypothesis that the organic acidic anions fixation by functional groups of the polymer matrix could facilitate the movement of the corresponding salt cations, providing a new perspective on improving the thermoelectric performance of traditional quasi-solid state i-TEs; on the other hand, the leather polymer matrix doped with $Na_3C_6H_5O_7$ (SC) had the highest Seebeck coefficient thus excellent thermoelectric property, indicating the anions of SC bearing suitable interaction with the leather polymer matrix thus maximizing cations movement and achieving the fastest speed.

Since the water content (*W*, %) of leather-based polymer matrix affected its nano-/micro-channel, thus influencing the rate of cation motion, in this section, the sample of LITE-$SC_{0.75\ M}$ with *W* of 70%, 60%, 50%, 40%, 30%, and 18% was prepared. As could be seen in Fig. 3a and Fig. S6a, similar to the influence of the SC concentration, with decreasing the *W*, the V_{oc} for all ΔT was increased first and then decreased. The Seebeck coefficients of them were further calculated to be 2.52, 4.23, 6.23, 5.98, 5.86, and 3.55 mV/K, respectively (Fig. 3b and Fig. S6b), *i.e.*, the sample of LITE-$SC_{0.75\ M}$ with *W* of 50% achieving the highest Seebeck coefficient (6.23 mV/K) thus excellent thermoelectric property. The higher water content of the leather-based polymer matrix, exemplified by 60% or even 70%, although more water could suffuse the leather nano-/micro-channels; meanwhile, it would increase the interaction between citrate and H_2O. The original hydrogen bonds among the citrate, *Emblic*, and leather collagen might be superseded by H_2O, manifesting as the citrate–H_2O anion aggregates exist in the leather collagen fibers' channels (Fig. 3c): first, it easily blocked the channels thus restricting the free movement of cations; second, anionic citrate–H_2O aggregates would weaken the total positive charge of free ions in the channels, thus reducing the rate of cations motion; Moreover, there might be an electrostatic interaction between cations and anionic citrate–H_2O, further detrimental to the free movement of cations. Therefore, the higher water content was disadvantageous to electrical energy output, manifesting as the Seebeck coefficient of LITE-$SC_{0.75\ M}$ with *W* of 70% dropping to 2.52 mV/K. However, when the water content of the leather-based polymer matrix was lower, the compact channel also restricted the free movement of cations, showing the Seebeck coefficient of LITE-$SC_{0.75\ M}$ with *W* of 18% down to 3.55 mV/K. Besides the reduced moisture on the migration efficiency of ions, the lower water content would include the alteration in ion concentration and the diminished activity of polar groups. These factors collectively contributed to the overall decline in ionic conductivity, which decreases as the water content decreases. As calculated based on electrochemical impedance spectroscopy, the ionic conductivity of LITE-$SC_{0.75\ M}$ was 2.89, 2.32, 0.63, and 0.11 S/m for *W* of 60%, 50%, 40%, and 30%, respectively (Fig. S7). In addition, the thermal conductivity of them was detected to be approximately 0.37, 0.32, 0.23, and 0.17 W/mK, respectively. This was normal as less water content of the matrix bearing better heat insulation performance. The good stability of their thermoelectric performance was proved by repeatedly turning on and turning off ΔT for 150 s, respectively (Fig. 3d). It was worth noting that for the sample with *W* of 50%, the V_{oc} dropped sharply when ΔT was turned off, while for other samples, especially for *W* of 30%, the V_{oc} dropped gently (Fig. 3d), the results of which also validated the restricted movement of cations for other water contents.

Except for the Seebeck coefficient (*S*), the figure of merit (ZT_i) and power factor (*PF*) are also used to comprehensively evaluate the thermoelectric performance of the i-TEs [35, 36]. The detailed values of them are illustrated in Fig. 3e and summarized in Table S2 obtained based on the calculation of equations $PF = S^2\sigma$ and $ZT_i = S^2\sigma T/\kappa$ (*T* is the absolute temperature (K), *S* represents the Seebeck coefficient (mV/K), σ denotes the ionic conductivity (S/m), and κ is the thermal

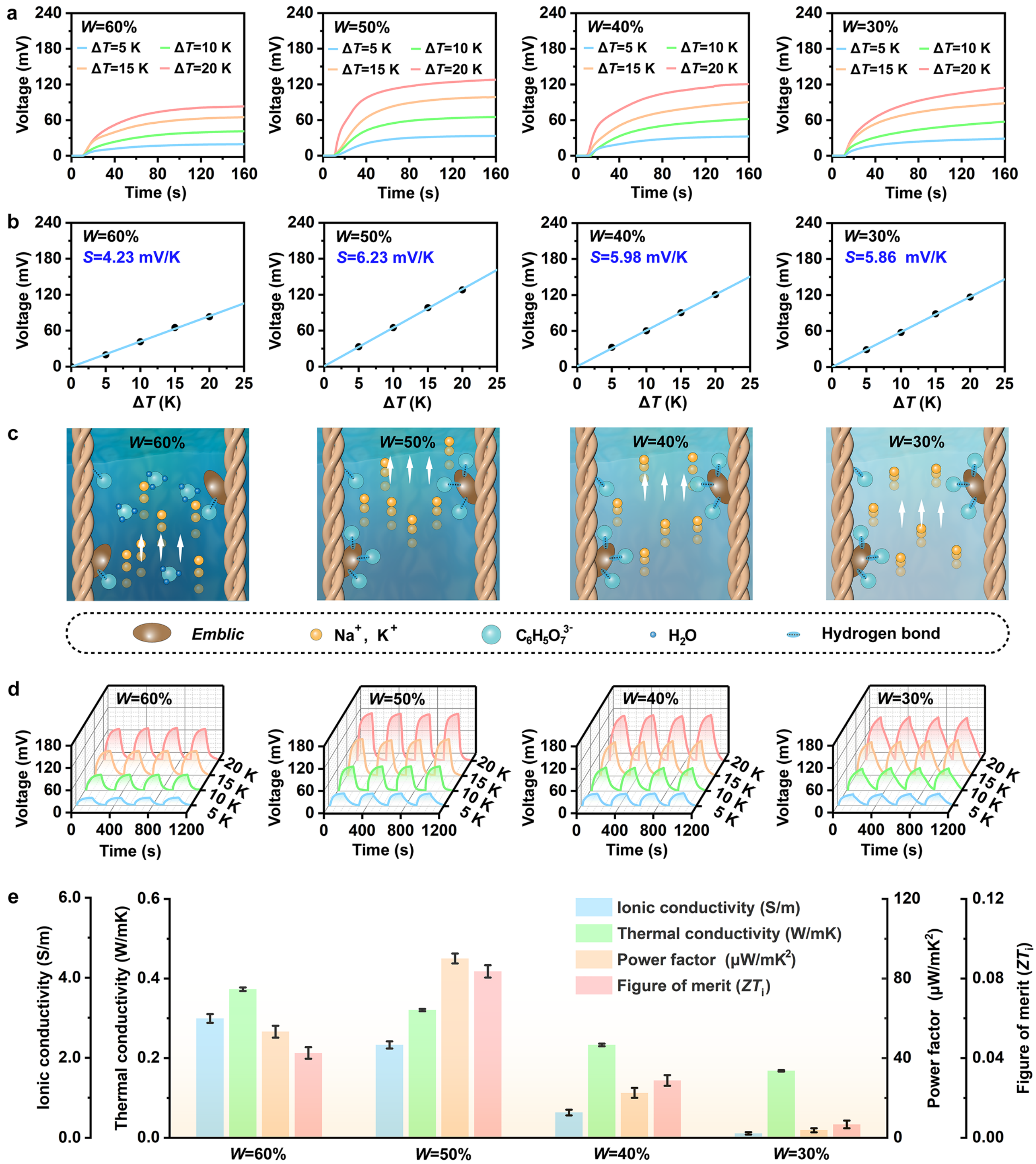

Fig. 3 **a** V_{oc} curves of LITE-SC$_{0.75\ M}$ with different water contents (W = 60%, 50%, 40%, 30%). **b** Seebeck coefficient of LITE-SC$_{0.75\ M}$ (W = X%) obtained from linear fitting of voltage and ΔT. **c** Schematic of the thermoelectric property of LITE-SC$_{0.75\ M}$ affected by water content. **d** V_{oc} curves of LITE-SC$_{0.75\ M}$ (W = X%) at turn-on/off cycles. **e** The values of ionic conductivity (S/m), thermal conductivity (W/mK), power factor (μW/mK^2), and figure of merit (ZT_i) of LITE-SC$_{0.75\ M}$ (W = X%)

conductivity (W/mK)). The PF and ZT_i of LITE-SC$_{0.75\ M}$ with W of 50% reached 90.05 μW/mK^2 and 0.084, approximately two times those of W of 60% (PF = 53.32 μW/mK^2 and ZT = 0.043), and even 24 and 12 times those of W of 30% (PF = 3.78 μW/mK^2 and ZT = 0.007), further indicating the excellent thermoelectric properties of the LITE-SC$_{0.75\ M}$ with W of 50%. The above results demonstrated the complex effect of water content on the nano-/micro-channels of the matrix and thus movement of ions. Therefore, it was important to control the water content of the matrix before application of i-TEs, exemplified by W of 50% suitable to the present leather-based polymer matrix.

3.2 Thermoelectric Performance of LITE-SC$_{0.75\ M}$

The thermoelectric performance of the optimized LITE-SC$_{0.75\ M}$ sample was evaluated. In this section and thereafter, the water content of LITE-SC$_{0.75\ M}$ was 50%. Figure 4a illustrates the current density–voltage curves for temperature difference (ΔT) of 5, 10, 15, and 20 K. The short-circuit current density (I_{scd}) and open-circuit voltage (V_{oc}) kept a linear relation, and both increased by increasing the ΔT. For example, the I_{scd} and V_{oc} were 30.1 mA·m^{-2} and 32.9 mV for ΔT of 5 K, while they were increased to 124.9 mA·m^{-2} and 127.4 mV for ΔT of 20 K. Unlike the I_{scd}–V_{oc} curves, the power density–voltage curves demonstrated that the power density increased first and then decreased with the increase of voltage (Fig. 4b). The maximum power density (P_{max}) for different ΔT is illustrated in Fig. 4c. It could be found that P_{max} was increased with the increase of ΔT. For ΔT of 20 K, the P_{max} was 4.35 mW·m^{-2}, 15 times that ΔT of 5 K (0.29 mW·m^{-2}). Moreover, the maximum specific power density ($P_{max}/(\Delta T)^2$) commonly used to evaluate the output performance of i-TEs was calculated and is demonstrated in Fig. 4d. It could be found that the P_{max} was approximately proportional to the $(\Delta T)^2$, which agreed with the theoretical result based on the equation of $P_{max} = (\sigma S^2/4d)\cdot(\Delta T)^2$ (d is the thickness of sample). In addition, $P_{max}/(\Delta T)^2$ could also indirectly evaluate the thermoelectric conversion capability of LITE-SC$_{0.75\ M}$ under different ΔT. When the ΔT increased from 5 through 10 to 20 K, the $P_{max}/(\Delta T)^2$ changed from 11.6 to 12.9 and then dropped to 10.8 μW·m^{-2}·K^{-2}, which indicated that the thermoelectric conversion capacity of LITE-SC$_{0.75\ M}$ decreased slightly with the increase of ΔT, showing more suitable for thermoelectric conversion of low-grade heat.

Subsequently, the influence of output voltage, current density, and power density with a different external load (R_L) under the condition of $\Delta T = 20$ K was investigated (Fig. 4e). It could be found that the average output voltage of LITE-SC$_{0.75\ M}$ increased with the increase of R_L and then gradually approached the maximum open-circuit voltage (Fig. 4f), while the average output current density showed an opposite trend, gradually approaching zero with the increase of R_L (Fig. 4g). Therefore, the average output power density eventually increased first and then decreased (Fig. 4h), reaching a maximum value of 2.54 mW·m^{-2} when the R_L was close to the internal resistance of LITE-SC$_{0.75\ M}$ (*i.e.*, 2×10^4 Ω). To further evaluate the stability of the output performance of the LITE-SC$_{0.75\ M}$ under long-term operation, an external resistor of 2×10^4 Ω was continuously tested for 7200 s under the condition of $\Delta T = 20$ K. It could be found that both output voltage (Fig. 4i) and current density (Fig. 4j) exhibited small attenuations over time, which was an acceptable phenomenon for the common i-TEs [37]. As for its corresponding power density curve (Fig. 4k), it appeared instantaneous power density of 4.20 mW·m^{-2} at the moment of the resistor turning on, which was close to the maximum power density under $\Delta T = 20$ K (P_{max}, 4.35 mW·m^{-2}, Fig. 4c). Moreover, based on the power density curve, the output energy density could be calculated to be 13.64 J·m^{-2} (Fig. 4k), which was comparable to or even higher than those of previously reported i-TEs [38–40]. Furthermore, the energy conversion efficiency (η_r, %) at the Carnot efficiency limit, an important parameter to assess the performance of i-TEs, was determined based on the equation of $\eta_r = P_{out}\cdot d\cdot T_{hot}/\kappa\cdot\Delta T^2$ (P_{out} is the output power density and T_{hot} is the temperature in the hot end). As demonstrated in Fig. 4l, the η_r reached up to 2.12% at the moment of the resistor turning on, higher than those of previously reported works [41, 42]. When further increased the time, the η_r stabilized around 1.00%. Finally, the primary thermoelectric performance indexes, Seebeck coefficient, and figure of merit, of present leather-based i-TEs were compared with other innovative works. As shown in Fig. 4m, the thermoelectric properties of LITE-XX (prepared using leather matrix doped with other salts) were superior to those of the vast majority of polymer-based i-TEs [43–52], indicating a good candidate for the leather-based matrix. Significantly, the thermoelectric performance of LITE-SC$_{0.75\ M}$ was further improved on the basis of LITE-XX, showing that trisodium citrate (SC) was the ideal organic acid salt. By employing the good candidate of polymer matrix conjugated with the ideal organic acid salt, the product LITE-SC$_{0.75\ M}$ achieved an excellent thermoelectric property.

3.3 Mechanism for the Electricity Induced by Temperature Difference of the Leather-Based Material

The above results indicated an excellent thermoelectric property of the LITE-SC$_{0.75\ M}$ sample. Herein, the mechanism for the electricity induced by temperature difference was tried to decipher by characterizing its surface morphology, microstructure, pore structure, elemental distribution, and chemical bonding. First, the macroscopic morphology showed that the leather grain exhibited a natural texture, while the flesh bearing the tightly woven structure of the leather fibers (Fig. 5a). The detailed grain surface could refer to its rendered ultra-depth 3D (Fig. 5b) and 2D (Fig. 5c) field images. The micron-level concave and convex structure (Fig. 5b) of the LITE-SC$_{0.75\ M}$ would increase the specific surface area of the interface between the electrode and the leather sample, thus enhancing the electrochemical reaction activity at the interface. The micron-level porous structure (Fig. 5c), generally used for the transportation of water and salts of animals, was significantly important for mobile ions transporting as i-TEs. Subsequently, the microstructure of LITE-SC$_{0.75\ M}$ was observed by SEM (Fig. 5d). It could be

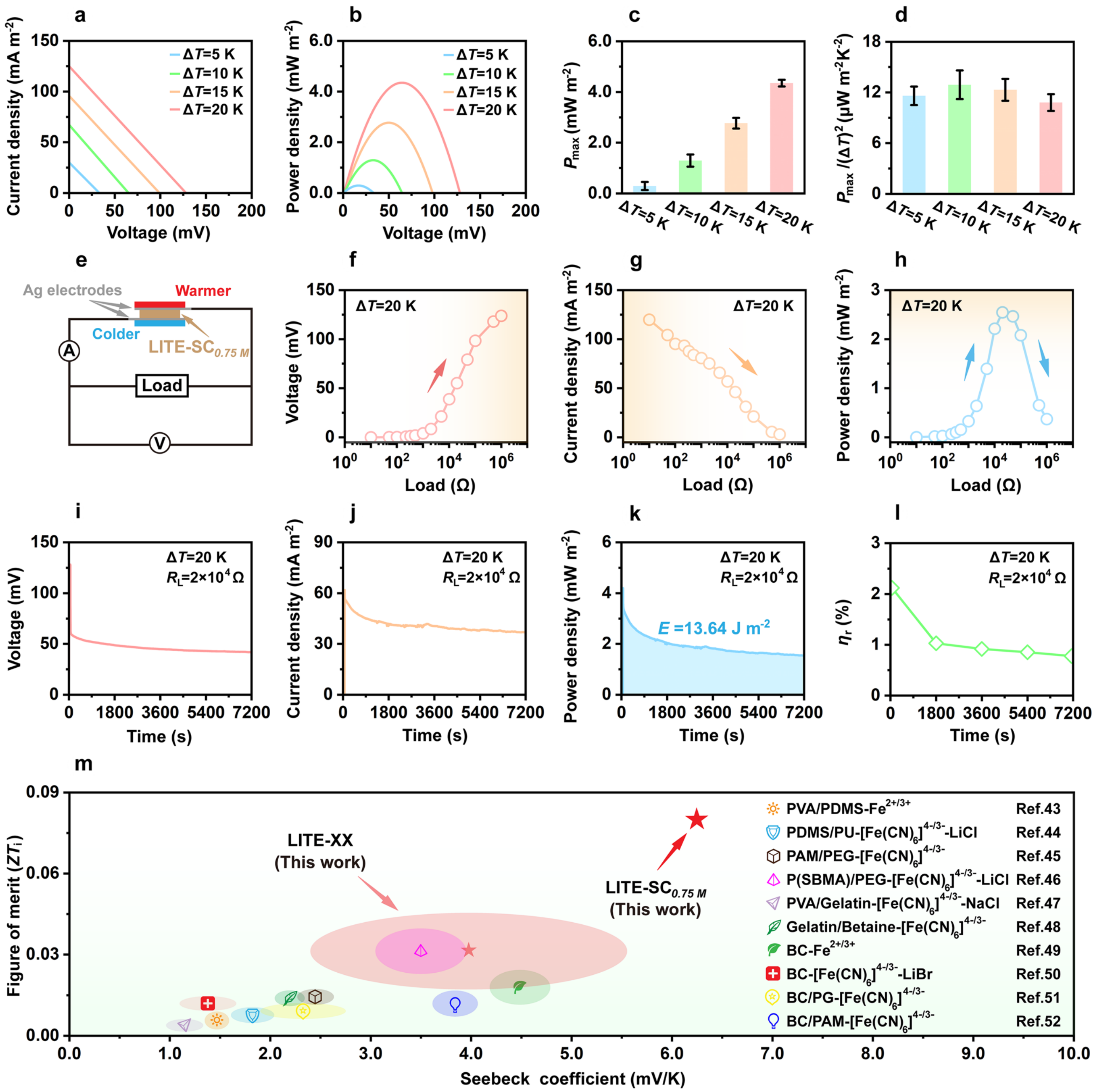

Fig. 4 **a** The current density–voltage plots, **b** the power density–voltage plots, **c** the maximum power density, and **d** the calculated maximum specific power density under different ΔT. **e** The schematic of the output voltage–current test with the external load. **f** The output voltage, **g** the output current density, and **h** the output power density with different external loads under $\Delta T = 20$ K. **i** The output voltage curve, **j** the output current density curve, **k** the output power density curve, **l** the η_r values at different time with the external load of 2×10^4 Ω under $\Delta T = 20$ K, and **m** the comparison of Seebeck coefficient (mV/K) and figure of merit (ZT_i) of LITE-SC$_{0.75\ M}$ with those of previously reported ionic thermoelectric materials

found that, on the one hand, similar to tree frog fins, octopus suction cups, and fish scales, leather collagen fibers demonstrated multilayered structures composed of complex networks of cross-woven, thus endowing the leather with excellent strength, elasticity, and toughness; on the other hand, collagen fibers were composed of an orderly arrangement of fibrils with thread-like structures, the gap of which formed nano-/micro-channels available for ions to move along specific pathways to improve transport efficiency, the greatest advantage of leather as a polymer matrix for i-TEs. The pore structure of LITE-SC$_{0.75\ M}$ was further analyzed by Mercury Piezometer technology (Fig. 5e). The cumulative pore

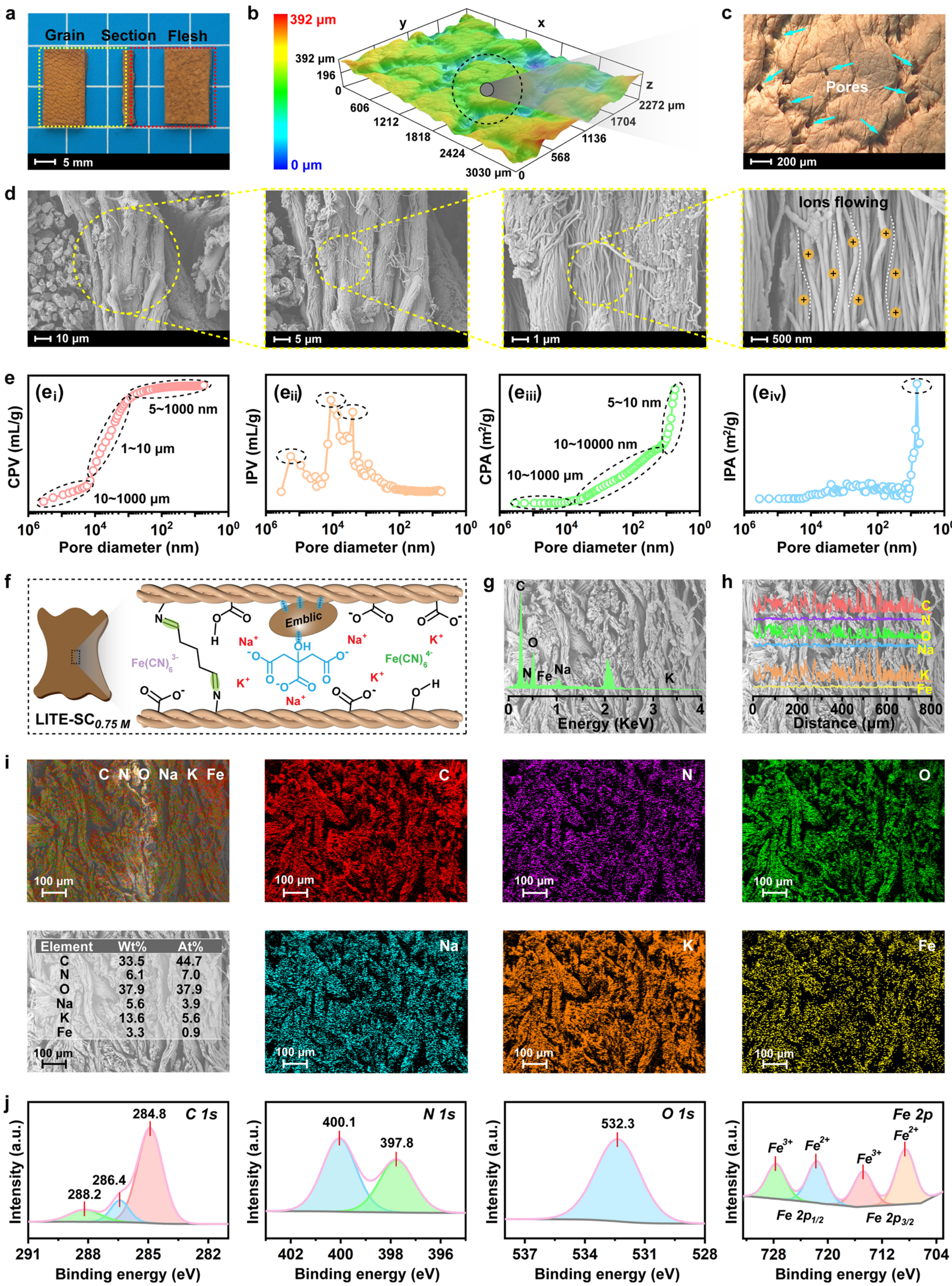

Fig. 5 The structure of LITE-SC$_{0.75\ M}$ sample: **a** the digital photos, **b** rendered ultra-depth 3D field image, **c** 2D image, **d** SEM images, **e** pore distribution curves, **f** chemical structure, **g** EDS spectrum, **h** EDS line scanning of N, O, C, Na, K, and Fe, **i** EDS map scanning and corresponding element content of different elements, and **j** XPS high-resolution spectra of *N 1 s*, *C 1 s*, *O 1 s*, and *Fe 2p*

volume (CPV) curve demonstrated three stages of leather pore diameters, attributing to the cross-weaving of collagen fibers (10 ~ 1000 μm), the orderly arrangement of secondary collagen fibers (1 ~ 10 μm), and the orderly arrangement of fibrils (5 ~ 1000 nm), respectively (Fig. 5e_i). The three peaks in the incremental pore volume (IPV) curve corresponded to the highest point of the increment volume and the transition point of the multistage structure of the collagen fibers (Fig. 5e_{ii}). Especially, the pore size of 5 ~ 1000 nm, the main channel for ion transport, occupied the largest volume number (Fig. 5e_i), demonstrating excellent ions transportability of LITE-$SC_{0.75\ M}$. The three stages of pore diameters in the cumulative pore area (CPA) curve were ascribed to the cross-weaving of collagen fibers (10 ~ 1000 μm), the orderly arrangement of fibrils (10 ~ 10,000 nm), and the triple-stranded helical structure (5 ~ 10 nm), respectively (Fig. 5e_{iii}). Significantly, the pore area increased rapidly at 5 ~ 10 nm, because the triple-stranded helices were the primary structure of the collagen fiber, coincident with the result of incremental pore area (IPA) curve that obtained the peak value in this pore diameter range (Fig. 5e_{iv}). Such result indicated that the triple-stranded helix bearing abundant functional groups of -NH_2, -OH, and -COOH occupied the largest specific surface area in the leather matrix, which provided a large number of hydrogen-bonding sites for the *Emblic* and citrate, thereby limiting the free movement of the anions.

Finally, the elemental composition/distribution/content and chemical bonding of LITE-$SC_{0.75\ M}$ (the proposed chemical structure, Fig. 5f) were characterized to further reveal the excellent performance of its electricity induced by temperature difference. As demonstrated in Fig. 5g, the EDS spectrum showed that LITE-$SC_{0.75\ M}$ mainly contained the elements of C, N, O, Na, K, and Fe. Among them, C, N, and O elements, accounting for the majority, were primarily from the leather matrix, Na from SC, and K and Fe from $K_3[Fe(CN)_6]/K_4[Fe(CN)_6]$. The uniform distribution of these elements along the collagen fibers could be confirmed by the EDS line (Fig. 5h) and map scanning (Fig. 5i). Significantly, the total weight percentage (wt%) of Na and K elements was relatively high (19.2 wt%), thus providing sufficient cations for the thermal migration under temperature difference. The chemical structure was analyzed by FTIR and XPS. The peaks of LITE-$SC_{0.75\ M}$ at 2926, 1643, and 1545 cm^{-1} in the FTIR spectrum (Fig. S8) were characteristic peaks of the leather collagen, corresponding to the stretching vibration of amide II with –CH_2–, the stretching vibration of amide I with C=O, and the stretching vibration of amide II with C–N and N–H bending vibrations, respectively. Compared to that of $Leather_{Emblic/GTA}$, the LITE-$SC_{0.75\ M}$ introduced new peaks at 2166 and 2038 cm^{-1} (Fig. S8), which were ascribed to the C≡N stretching vibration of $[Fe(CN)_6]^{3-}/[Fe(CN)_6]^{4-}$, indicating the $[Fe(CN)_6]^{3-}$ and $[Fe(CN)_6]^{4-}$ stabilized inside the leather matrix in their oxidized and reduced states, respectively [48]. Figure 5j showed the high-resolution XPS spectra of *C 1 s*, *N 1 s*, *O 1 s*, and *Fe 2p*, respectively. The peaks at 284.8, 286.4, and 288.2 eV in the *C 1 s* spectrum were attributed to C–C/C–H, C–O, and C=O in the collagen, respectively; the peaks at 397.8 and 400.1 eV in the *N 1 s* spectrum were due to N–H/C=N and C–N in the collagen, respectively; the broad peak at 532.3 eV in the *O 1 s* spectrum corresponded to N–O, C=O, C–O, and O–H in the collagen, *Emblic,* and SC; and the double peaks of $Fe\ 2p_{1/2}$ and $Fe\ 2p_{3/2}$ in the *Fe 2p* spectrum indicated the presence of both Fe^{3+} and Fe^{2+}, further demonstrating that $[Fe(CN)_6]^{3-}$ and $[Fe(CN)_6]^{4-}$ maintained their respective oxidized and reduced states, which was necessary for exerting the thermogalvanic effect. Based on the above results, the chemical structure of LITE-$SC_{0.75\ M}$ was proposed as displayed in Fig. 5f. The *Emblic*, GTA, and citrate anions were combined with the collagen fibers by hydrogen and covalent bonds, whereas Na^+/K^+ and $[Fe(CN)_6]^{3-/4-}$ synergistically achieved temperature difference-dependent electricity generation by thermodiffusion and thermogalvanic, respectively.

3.4 Functional Application of LITE-$SC_{0.75\ M}$ as an Automatic Anti-scalding Detector

To ensure the feasibility of LITE-$SC_{0.75\ M}$ in practical functional applications, the properties, including thermal stability, mechanical properties, durability, and cost, were assessed first. The thermal stability was evaluated through shrinkage temperature (T_s), thermogravimetric analysis (TG/DTG), and thermal degradation temperature (T_d). As shown in Fig. 6a, the T_s of LITE-$SC_{0.75\ M}$ reached 90.4 °C, comparable with or even higher than those of common leather-based materials [53–55], indicating good hydrothermal stability. Figure 6b, c shows its TG and DTG curves. The weight loss of LITE-$SC_{0.75\ M}$ in the range of 30–110 °C was primarily due to free water evaporation; the weight loss with a fast speed in the range of 280–400 °C was mainly attributed to the degradation of collagen fibers; while the weight loss after 400 °C was ascribed to decomposition and sublimation of carbonization products. The thermal decomposition of LITE-$SC_{0.75\ M}$ approximately began at 300 °C and peaked at 326 °C (Fig. 6c), demonstrating a heat-resistant characteristic suitable for applications in automatic anti-scalding alarm systems. Moreover, the thermal degradation temperature, T_d, of LITE-$SC_{0.75\ M}$ was 120 °C (Fig. 6d), indicating its suitability for storage and transportation at high temperatures. Therefore, thermal analysis confirmed that LITE-$SC_{0.75\ M}$ possessed a wide operating temperature range and a high heat-resistance limit, making it suitable for thermoelectric conversion applications in environments with large temperature variations.

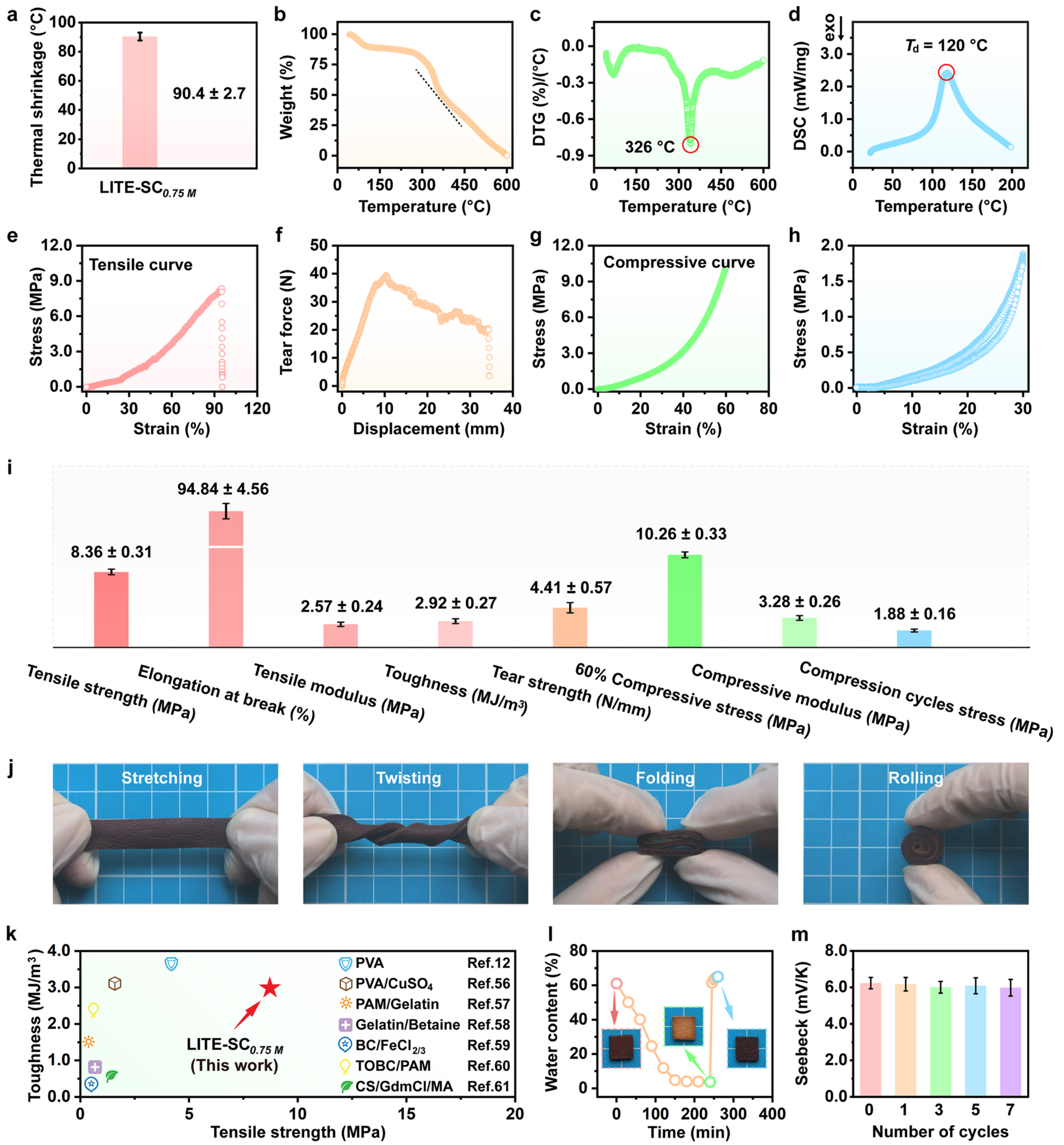

Fig. 6 **a** Shrinkage temperature (T_s). **b** TG curve. **c** DTG curve. **d** DSC curve. **e** Tensile stress–strain curve. **f** Tear force–displacement curve. **g** Compressive stress–strain curve. **h** Compression cycles curves. **i** The values of mechanical property calculated by the above curves. **j** Digital photos for stretching, twisting, folding, and rolling. **k** The comparison of the tensile strength and toughness of LITE-$SC_{0.75\ M}$ with those of previously reported ionic thermoelectric materials. **l** Water content curve over time by drying-restoring test. **m** Seebeck coefficients after different drying-recovery cycles of LITE-$SC_{0.75\ M}$ sample

Subsequently, the comprehensive mechanical performance of LITE-$SC_{0.75\ M}$ was assessed by tensile, tearing, compression, and circulation tests. According to its tensile stress–strain curve (Fig. 6e), the common indexes including tensile strength, elongation at break, tensile modulus, and toughness were calculated. For instance, the tensile strength and toughness of LITE-$SC_{0.75\ M}$ were 8.36 MPa and 2.92 MJ/m^3. The strong tensile modulus (2.57 MPa) and high elongation at break (94.84%) could also be proved by its easily lifting 5 L water (Fig. S9). Moreover, based on its tear force–displacement curve (Fig. 6f), the tear strength could be calculated to be 4.41 N/mm, indicating that it was capable of withstanding considerable external tearing stress. Also, LITE-$SC_{0.75\ M}$ exhibited excellent compressive properties. As demonstrated in Fig. 6g, the compression stress and modulus could be calculated to be 10.26 MPa and 3.28 MPa at 60% strain. Ten cyclic compressions at 30% strain showed good fatigue resistance and elasticity (Fig. 6h). Such excellent flexibility (Fig. 6i) of LITE-$SC_{0.75\ M}$ could also be confirmed by optionally stretching, twisting, folding, and rolling it (Fig. 6j), which was thus allowed to design into any shape of the thermoelectric devices. Combined with its excellent mechanical properties (exemplified by its tensile strength and toughness significantly higher than those of previously reported polymer-based thermoelectric materials (Fig. 6k) [12, 56–61]), LITE-$SC_{0.75\ M}$ demonstrated a good candidate for a variety of complex application scenarios.

The durability of LITE-$SC_{0.75\ M}$ was evaluated through drying-recovery and natural placement tests. As depicted in Fig. 6l, LITE-$SC_{0.75\ M}$ could completely lose moisture in an oven at 60 °C. Afterward, it quickly absorbed water and recovered to its initial moisture content within 5 min after re-immersion in a solution with the same concentration of ions. After seven cycles of drying and recovery, it could be found that the Seebeck coefficients did not exhibit an obvious decrease (Fig. 6m). Moreover, by placing the LITE-$SC_{0.75\ M}$ in a natural environment for 10 days, no noticeable change was observed in appearance (Fig. S10). However, a green color appeared on the LITE-$SC_{0\ M}$ without the addition of SC (Fig. S10). Such phenomenon was ascribed to the reaction between $K_3[Fe(CN)_6]$ and $K_4[Fe(CN)_6]$ forming $Fe_3[Fe(CN)_6]_2$, which would undoubtedly reduce its thermoelectric properties. Fortunately, this dilemma could be avoided in the LITE-$SC_{0.75\ M}$ sample due to the buffering effect of SC thus inhibiting the above reaction. Moreover, the cost of the fabricated leather-based LITE-$SC_{0.75\ M}$ was approximately 164.65 CNY/kg (Table S3), significantly lower than previously reported i-TEs such as PVDF-HFP, PEDOT:PSS, PVA-PDMS, and BC (Table S4). The above results indicated excellent reversibility, durability, stability, and low cost of LITE-$SC_{0.75\ M}$, establishing a solid foundation for its long-term application in natural environments.

In the last section, the potential application of LITE-$SC_{0.75\ M}$ as an automatic anti-scalding detector was evaluated. As shown in Fig. 7a, b_i, first, ten LITE-$SC_{0.75\ M}$ units were connected in series using Cu electrodes and packaged flexibly with BOPP film to create a simple device, which demonstrated excellent bendability (Fig. 7b_{ii}) and foldability (Fig. 7b_{iii}), providing both flexibility and portability for practical use. Such a device could generate a high V_{oc} of 1.26 V under ΔT of 20 K (Fig. 7c), approximately ten times that of an individual one (127.4 mV, Fig. 4a), further demonstrating the potential application of LITE-$SC_{0.75\ M}$ in integrated thermal harvesting. Similar to those common i-TEs, the generated V_{oc} could almost completely charge capacitors of different capacities, with increasing charging time for increased capacity of the capacitors (Fig. 7d). For example, the device could successfully charge a 1000 μF capacitor to 1.23 V within 3600 s, showing its excellent energy conversion efficiency and energy storage capability. Moreover, such devices could harvest low-grade heat from various environments, such as laptops, smartphones, forearms, and 10 °C cold water (Fig. S11), highlighting its diverse applications. The varied V_{oc} was due to their different contribution of ΔT (Fig. 7e-h). For instance, the device could generate V_{oc} of 0.87 V by harvesting low-grade heat from an operational laptop (Fig. 7e), while a reverse V_{oc} of 0.55 V from a cold water cup (10 °C) at ambient temperature (28 °C) (Fig. 7h). On the other hand, the good thermal sensing as well as thermoelectric power generation ability were further confirmed by the fixation of LITE-$SC_{0.75\ M}$ unit on the five fingers of a glove (Fig. 7i, Fig. S12). Such sensing glove could generate a reverse V_{oc} when grasped in cold water (Fig. 7j), while a forward V_{oc} when grasped in hot water (Fig. 7k). The cyclic experiment of grasping hot water validated the sensing stability during use (Fig. 7l). Considering its excellent low-grade heat harvesting and thermal sensing capabilities, an automatic anti-scalding detector based on LITE-$SC_{0.75\ M}$ was designed (Fig. S13). As illustrated in Fig. 7m, n, the anti-scalding detector could monitor different levels of risk such as low (electric blanket), medium (hot-water bag), high (electric heater), and extreme (kettle containing boiling water) risks, manifesting as inputting the stronger thermal sources outputting the higher V_{oc}. When the temperature reached the set danger threshold, the generated V_{oc} could drive the signal transmitter to alert the user avoiding the occurrence of scalding accidents. Therefore, installing the leather-based automatic anti-scalding alarm detector on the skin or clothing of children or the elderly enabled the timely detection of its contact with high-temperature objects and meanwhile risk assessment, effectively reducing property loss, injuries, or even casualties (Fig. 7n). Such robust and efficient natural leather-based ionic thermoelectric detectors with energy-autonomous performance showed great promise in automatic anti-scalding alarm.

4 Conclusions

In this work, a robust and efficient natural leather-based ionic thermoelectric material (LITE-$SC_{0.75\ M}$) was successfully fabricated by introducing SC and $K_3[Fe(CN)_6]$/$K_4[Fe(CN)_6]$ into the $Leather_{Emblic/GTA}$. By optimizing the mobile ions of SC and channels of leather collagen fibers matrix as well as

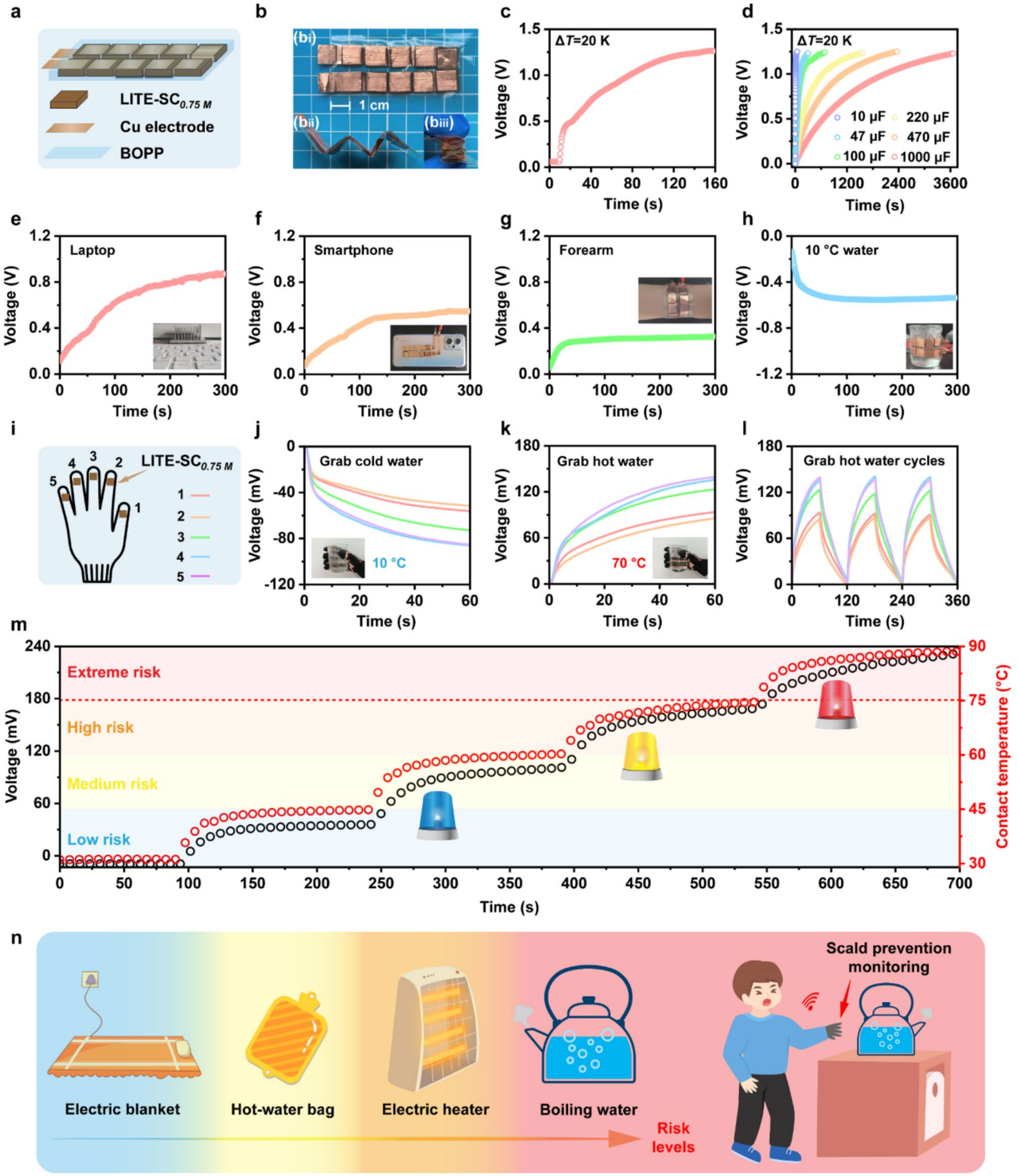

Fig. 7 **a** Schematic of ten LITE-SC$_{0.75\ M}$ in series. **b** Photograph of (b_i) ten LITE-SC$_{0.75\ M}$ in series, (b_{ii}) multiple bending, and (b_{iii}) folding. **c** V_{oc} curve of ten LITE-SC$_{0.75\ M}$ in series. **d** Charging curves for external capacitors of different capacities. The V_{oc} curve of ten LITE-SC$_{0.75\ M}$ in series for low-grade heat collection from **e** laptop, **f** smartphones, **g** forearm, and **h** 10 °C water. **i** Schematic of heat-sensing glove. The V_{oc} curves of the glove grabbing **j** cold water, **k** hot water, and **l** hot-water cycles. **m** The V_{oc} and contact temperature curves of the automatic anti-scalding device under different risk levels. **n** The risk levels and the schematic of application of the leather-based ionic thermoelectricity for scald prevention monitoring

conjugating with the thermogalvanic effect of $K_3[Fe(CN)_6]$/$K_4[Fe(CN)_6]$, the LITE-SC$_{0.75\ M}$ showed excellent thermoelectric performance, including a Seebeck coefficient up to 6.23 mV/K, a figure of merit of 0.084, and an energy conversion efficiency of 2.12%. The Leather$_{Emblic/\mathrm{GTA}}$ matrix providing suitable nano-/micro-channels together with endowing abundant interface structure as well as SC enabling the fast moving of cations was the mechanism for the efficient electricity induced by temperature difference of the LITE-SC$_{0.75\ M}$. The assembled devices with outstanding capabilities for low-grade heat harvesting and thermal sensing showed great promise in automatic anti-scalding alarms. Since the leather-based ionic thermoelectric detectors with energy-autonomous performance were endowed with excellent mechanical performance, thermal stability, durability, and inexpensive, they would exhibit long service life and can also realize application in multiple scenarios and extreme environments.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s42765-025-00529-6.

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Data Availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of Interest The authors declare that they have no conflict of interest.

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